

CRV lite

Constrained random verification means you declare what can vary

Rand, constraints, and randomize

- Mark fields with rand so the solver may assign them
- A constraint block carves the legal region
- Randomize returns one if a legal sample was found
- Seed controls reproducibility, same seed, same stream when you replay
- Conflict presets show overlapping hard constraints with an empty intersection
- Real CRV uses a full solver

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

Constraint / random lite

Dice-level CRV literacy: rand fields, a simple constraint, randomize() success/fail, and a seed. Not a full solver. Starter: **data inside {[0:15]}** with one legal roll.

Starter example: data inside {[0:15]}, seed 42, one legal roll already in the histogram.

Load starter example

Challenges 0 / 22 cleared

Quiz: rand: A rand field is...

☐ a variable the randomize() solver may assign

☐ always a wire in the DUT

☐ only legal in always_comb

☐ a synthesis keep attribute

Show hint

Check

Next

Idle

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Core ideas

rand

Fields the solver may assign.

constraint

Legal region over those fields.

randomize()

1 = ok sample; 0 = unsat / fail.

seed

Same seed → same dice stream.

Lab

CONSTRAINT PRESET

SEED

Range {[0:15]}

42

Load preset

randomize()

Roll x10

Apply seed

CRV lite lab starter

Real UVM literacy

```
ws1 - module12-crv-lite/examples/crv-sketch

# Track A - real Unix
# real shell session (Track A - CRV lite / randomize)

$ pwd
/mt/d/proj/designs/learning/courses/learn_uvm2017/module12-crv-lite

$ ls examples/crv-sketch/
randomize.txt

$ cat examples/crv-sketch/randomize.txt
Constrained random verification sketch (literacy)

Pattern on a class or seq_item:
class MyTxn extends uvm_sequence_item;
  rand bit [7:0] data;
  rand bit [15:0] addr;

  constraint c_range {
    data inside {[0:15]};
  }

  constraint c_align {
    addr % 4 == 0;
  }
endclass

MyTxn txn = MyTxn::type_id::create("txn");
if (!txn.randomize())
  `uvm_fatal("RAND", "randomize failed")
// txn.data, txn.addr now hold legal sample values

Steps:
1) declare rand fields
2) write constraint { ... } for legal region
3) call randomize() + 1 = ok, 0 = fail (constr / solver error)
4) use sampled values in sequence / driver

Seed (reproducibility):
initial begin
  void'(std::randomize(seed)); // or +SEED plusarg
end
// same seed + same constraints + same stream (debug flakiness)

Conflict example (randomize + 0):
constraint c_a { x inside {[0:3]}; }
constraint c_b { x inside {[12:15]}; }
// empty intersection + no legal x

Common mistakes:
conflicting hard constraints
forgetting to check randomize() return value
assuming randomize success = coverage done
no seed when debugging intermittent failures

UVM tip-in:
seq_item.randomize() before start_item
plus SEED plusarg from module 11 for replay

$ grep -n "rand \|constraint\|randomize" ../../learn_uvm2017_sv_verilator/module8/examples/random_utils/random_utils.sv
1 head -12
264: * RandomTransaction: Transaction with random constraints
269: *   - Random fields with constraints
274:   rand logic [7:0] data; // Data (8-bit, constrained)
275:   rand logic [15:0] address; // Address (16-bit, constrained)
276:   rand int delay; // Delay (constrained)
286:   constraint valid_data {
290:     constraint valid_address {
294:     constraint valid_delay {
379:       void'(txn.randomize());
```

Real shell — CRV sketch

Pitfalls to watch

- Do not treat randomize success as coverage done, you still need checks and covergroups
- Do not write conflicting hard constraints and expect the solver to pick a side
- Do not forget seed when debugging a flaky failure, you need replay
- Soft constraints exist in full SV but this lab focuses on hard constraints first
- And remember: the browser histogram is discrete; real solvers search huge spaces

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, load conflict and explain why randomize returns zero
- On real UVM, sketch one rand transaction with two constraints
- When you are ready, take the short quiz, then continue to scoreboards in the next module